**Table 1:** Related studies: reported targets, data units and evaluation approaches.

| Study | Response | Data unit | Evaluation |
|---|---|---|---|
| Wisniewski et al. (2020) | Next-hour signed-position interval | One market maker's hourly FX positions | Pinball and interval scores; coverage tests |
| Boudoukh et al. (2004) | Worst cumulative portfolio-return quantile | Fixed portfolio under a lognormal diffusion | MaxVaR/VaR ratios; discrete-monitoring simulation |
| van den Hoogen (2017) | Conditional quantile of cumulative payment-deficit peak | ABN AMRO transaction accounts | Coverage and independence backtests |
| This study | Sampled absolute-position peak | FX positions; weekly position aggregates | Pinball loss, breaches, model confidence sets and costs |

**Table 2:** Data and matched six-record peak summaries.

*Panel A. Sources, aligned samples and existing split dates*

| Data | Frequency | Series / records | Unit | Sample | Validation | Evaluation |
|---|---|---|---|---|---|---|
| FX, hourly | Hourly | 1 / 18,576 | USD mn | 2014-02 to 2017-01 | 2015-10 to 2015-12 | 2016-01 to 2017-01 |
| FX, daily | Day end | 1 / 930 | USD mn | 2014-02 to 2017-01 | 2015-02 to 2015-04 | 2015-05 to 2016-12 |
| Treasury bills | Weekly | 1 / 706 | USD mn | 2013-04 to 2026-10 | 2019-07 to 2020-07 | 2021-01 to 2026-04 |
| NY Fed panel | Weekly | 12 / 706 | USD mn | 2013-04 to 2026-10 | 2019-07 to 2020-07 | 2021-01 to 2026-04 |
| CFTC panel | Weekly | 17 / 877 | Contracts | 2010-01 to 2026-10 | 2020-08 to 2021-08 | 2021-09 to 2026-09 |

*Panel B. Within-series mean, sample SD and empirical 95th percentile*

| | | $\lvert N \rvert$ | | | Peak $Y$ | | | Headroom $Z$ | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Data | Unit | Mean | SD | 95th | Mean | SD | 95th | Mean | SD | 95th |
| FX, hourly | USD mn | 8.92 | 10.2 | 27.6 | 11.1 | 11.7 | 30.2 | 2.15 | 5.92 | 7.96 |
| FX, daily | USD mn | 9.00 | 10.5 | 27.5 | 17.9 | 16.6 | 41.0 | 8.91 | 14.8 | 24.9 |
| Treasury bills | USD mn | 39,500 | 27,200 | 92,600 | 55,300 | 30,400 | 108,000 | 15,800 | 16,000 | 50,000 |
| NY Fed panel | USD mn | 12,400 | 8,340 | 33,300 | 16,000 | 9,040 | 37,400 | 3,750 | 3,560 | 10,500 |
| CFTC panel | Contracts | 60,200 | 46,000 | 148,000 | 81,800 | 45,600 | 169,000 | 19,900 | 20,900 | 60,600 |

*Notes.* Records are per series; complete $H = 6$ peak counts are 18,570, 924, 700, 700 and 871, respectively, including the issue anchor. Earlier hourly FX validation spans 2015-04 to 2015-07 and evaluation spans 2015-07 to 2015-10; daily FX uses each nonempty source-naive day's last record, and both FX periods had been inspected. SD uses $m - 1$ and the empirical 95th percentile has rank $\lceil 95m/100 \rceil$; dollar values are in millions, panel entries are separate within-series medians without exposure pooling, and CFTC retains heterogeneous native contract counts.

**Table 3:** Forecast definitions and executed menus. Codes occur here to connect with the result files. Matched histories and update schedules make the reference explicit.

| Method | Forecast definition | Executed menu / setting |
|---|---|---|
| Historical headroom (F) | Current floor plus a quantile of completed peak headrooms | Original FX histories {720,2880}; weekly {52,260} |
| Translated path (S) | Translate completed signed paths to the current position before taking their peaks | Same completed-label pools; fixed-window and rolling-window controls |
| Historical peak (K) | Historical absolute-peak quantile projected to the current floor | Same histories and update controls |
| Quantile regression (QR) | Training-scaled headroom regression with ridge slopes and an unpenalized intercept | Original three-point menu; initial panel five-point expansion; six-point menu $\{0.001, 0.01, 0.1, 1, 10, \infty\}$ |
| Local quantiles (KNN) | Neighbor headroom quantile, with a separate increment-scale variant | Training-only transforms and fixed neighbor menus; Supplement Sections S8 and S23 |
| Four-feature regression (QR4) | Current signed/magnitude positions and short/long increment RMS | Original five-point menu; six-point paired comparison below |
| Endpoint plug-in (L) | Floor plus a high absolute-displacement endpoint quantile | Proposition 2; empirical plug-in evaluated separately |
| Further benchmarks | Autoregressive replay; filtered paths; peak-tail fit; quantile forest | Existing bounded specifications; results and complete failures linked from Supplement Sections S7 and S34 |
| Coverage and combinations | Delayed projected adaptive conformal inference (ACI) wrappers; symmetric absolute value (SAV) headroom recursion; fixed forecast combinations | Original-source wrapper/SAV inventory and broader combination inventory are separate |

Training histories count completed labels. Fixed-window models refit every 24 FX records or 13 weekly records. Quantile and normalization conventions are detailed in Supplement Sections S8 and S23; this table retains the executed menus.

**Table 4:** Long-history short-horizon FX regression under two references and periods. Negative percentages favor regression. Matched gains recur while the rolling-reference direction changes.

| Period / procedure | Reference | Loss change (%) | Issues | Strict breaches (regression / reference) |
|---|---|---|---|---|
| Later / original main comparison | Shorter-history rolling headroom | -12.66 | 6625 | 376 / 332 |
| Later / matched | Same-history fixed-window headroom | -14.77 | 6625 | 376 / 360 |
| Earlier / original structure | Shorter-history rolling headroom | +3.17 | 1440 | See complete earlier-period tables |
| Earlier / matched | Same-history fixed-window headroom | -16.62 | 1440 | 20 / 16 |
| Later / coverage-wrapped | Same-history wrapped headroom | -15.00 | 6625 | 329 / 328 |

Regression uses history 2880 and horizon 6; the original rolling reference uses 720. The earlier matched comparison is the original three-penalty regression, not the later one-standard-error selection rule. The wrapper comparison is a separate finite projected delayed ACI-style exercise.

**Table 5:** Weekly evidence across the original main comparison, added panel and earlier-period comparisons. Counts refer to related tasks rather than independent dealers. Stronger regularization narrows losses while leaving substantial failures.

| Comparison set | Procedure | Loss change / median (%) | Higher-loss cells |
|---|---|---|---|
| Original Treasury-bill main comparison | Original regression, long history | +50.79 | Original main comparison |
| Earlier Treasury-bill short-window version | Short-history short-window version | +144.30 | Different history |
| Additional panel | Original full regression | +92.63 | 45 / 48 |
| Additional panel | Four-feature regression | +9.91 | 37 / 48 |
| Additional panel | Full regression, expanded penalties | +9.22 | 34 / 48 |
| Additional panel | Ordinary empirical headroom quantile | +0.50 | 30 / 48 |
| Earlier weekly groups | Validation-minimum expanded rule | +4.15 | 26 groups |
| Earlier weekly groups | One-standard-error rule | +5.37 | 26 groups |

The original bill main comparison has 54 versus 28 strict breaches among 277 issues. Earlier short-history bill results use a different history; the one-standard-error worst excess is +181.39%, and all its prewritten success criteria remain unmet. Supplement Sections S11–S15 list all panel and selection results (Tables S9–S13).

**Table 6:** CFTC dealer-category futures: relative loss versus matched historical headroom (%). All 17 selected literal contract codes are retained.

| Contract | History 52 | | | | | History 260 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| code | Intercept only | Four-feature | Full regression | Calibrated history | Calibrated regression | Intercept only | Four-feature | Full regression | Calibrated history | Calibrated regression |
| 044601 | 5.26 | 128.00 | 41.38 | 15.01 | 18.80 | 0.51 | 7.82 | 43.88 | 13.87 | 42.41 |
| 043602 | 3.37 | 9.35 | 3.37 | 37.79 | 24.64 | 2.30 | 8.95 | 20.97 | -11.22 | -12.64 |
| 042601 | 4.64 | 9.48 | 35.65 | 26.69 | 41.21 | -0.15 | 74.23 | 16.63 | 1.46 | 9.39 |
| 045601 | -2.04 | 0.31 | -2.04 | 21.83 | 6.14 | 1.85 | 1.85 | 1.85 | 2.40 | 2.58 |
| 020601 | 4.77 | 79.72 | 111.12 | 25.66 | -5.34 | 0.92 | 43.38 | 29.70 | -15.12 | -12.32 |
| 099741 | 0.69 | 13.97 | 14.56 | 77.33 | -13.28 | 0.22 | 0.22 | 0.22 | 27.72 | 27.73 |
| 1170E1 | 6.42 | 6.82 | 46.30 | -0.22 | 12.77 | 0.92 | -28.44 | -16.22 | -1.12 | 1.00 |
| 097741 | -0.83 | 23.76 | 68.71 | 29.77 | 33.79 | 0.53 | -6.54 | 14.69 | -0.41 | -5.43 |
| 096742 | 2.93 | 3.41 | 6.13 | 49.54 | 30.46 | 5.30 | 4.82 | 62.68 | 37.12 | -0.49 |
| 095741 | 7.39 | 19.55 | 19.33 | 41.27 | 42.98 | -0.70 | 2.13 | 10.35 | 19.30 | 55.39 |
| 090741 | 4.03 | 10.28 | 4.03 | 15.58 | 13.13 | 3.28 | -5.35 | -3.60 | -2.02 | -5.66 |
| 232741 | 0.11 | 123.33 | 37.53 | 63.84 | 36.35 | 1.94 | -3.35 | 7.68 | -0.65 | -3.86 |
| 33874A | 6.95 | 88.57 | 38.30 | 30.15 | 44.07 | -2.38 | -3.21 | -4.78 | 6.98 | 9.52 |
| 240743 | -3.04 | -7.32 | 183.12 | 33.14 | 49.77 | -2.28 | -2.28 | -2.28 | 32.91 | 26.37 |
| 092741 | 0.54 | 0.54 | 0.54 | -1.74 | -0.68 | 0.17 | -6.98 | 26.22 | 5.58 | 13.94 |
| 098662 | -0.44 | -0.44 | 0.90 | 44.89 | 38.90 | 2.45 | -39.44 | 2.45 | 15.12 | 20.45 |
| 112741 | 11.06 | 38.76 | 10.05 | 33.72 | 24.10 | 2.24 | 2.82 | 80.98 | 8.36 | 16.37 |

Entries are percentage loss differences from matched corrected historical headroom; positive values mean higher loss, and unavailable entries retain failed or incomplete comparisons. The pre-specified conditions require historical superiority in at least 60% of all 34 comparisons, full-regression medians within ten percent at both histories, and a strictly smaller four-feature median with longer history. Full-precision results determine the comparisons, with all scores, breaches, selections and failures retained in Supplement Section S30.

**Table 7:** Full-regression excess loss versus matched historical headroom under longer validation. Entries are source medians in percent; the final column counts selections of the intercept-only limit.

| Reporting system | Horizon | History | Series | Validation 52 | Validation 156 | Limit selections |
|---|---|---|---|---|---|---|
| CFTC | 6 | 52 | 17 | +19.33 | +4.03 | 4 to 10 |
| CFTC | 6 | 260 | 17 | +10.35 | +0.53 | 4 to 5 |
| New York Fed | 6 | 52 | 13 | +6.68 | -1.13 | 6 to 9 |
| New York Fed | 24 | 52 | 13 | +40.47 | +5.02 | 3 to 9 |

Lengths count weekly records; evaluation dates remain unchanged. New York Fed horizon six also moves the validation endpoint and refit phase; across-arm medians are not typical paired changes. Supplement Section S38 gives individual series, four-feature results and the complete figure.

**Table 8:** Full-regression relative excess loss (%) at alternative quantile levels. Positive values favor matched historical headroom; NA retains any failed required comparison.

| Source / training history | 90% | 99% |
|---|---|---|
| CFTC, 52 weeks | +3.45 | +15.05 |
| CFTC, 260 weeks | +1.58 | NA |
| FX, original period | -10.15 | NA |
| FX, earlier period | -17.85 | NA |
| Treasury bills | +0.67 | +44.22 |

Contract rows give full-group medians; histories are 2,880 hours for FX and 260 weeks for Treasury. The rejected post-failure repair leaves original NA entries unchanged (Supplement Section S38).